

# Mixed Signal Processing With Analogue to Digital Convertors ADCs Notes

Bedded Systems 1

Semester 5

Page 1

Este se cumple por las props de la AND  $\rightarrow$  AND

Es loco!!!  $\rightarrow$

|      |      |      |      |
|------|------|------|------|
| 1011 | 0010 | 1111 | 0010 |
| 1111 | 1111 | 1111 | 1101 |
| 1011 | 0010 | 1111 | 0000 |

## Mixed signal processing using the MSP430-G2553 – use of analog on-chip devices.

Each version of the MSP430 contains a range of analogue circuitry –comparators, sensors, amplifiers, ADCs, DACs etc, each of which may be controlled in MSP430 user-software by configuring appropriate internal registers.

The G2553 incorporates one 10-bit Analogue to Digital Convertor – ADC10

The ADC has a 10-bit output, as the name suggests – its output ranges from 0000000000B (0000H) to 1111111111B (03FFH), or from 0 to 1023 decimal.

- The ADC converts an analog input to its 10-bit digital representation and stores the result in the special **ADC10MEM** register. The convertor uses two programmable/selectable voltage levels ( $V_{+ref}$  and  $V_{-ref}$ ) to define the upper and lower limits of the conversion i.e the reference voltage. Often the lower limit ( $V_{-ref}$ ), is ground and the upper limit is referred to simply as  $V_{REF}$ . Then the convertor converts input voltages in the range 0V to  $V_{REF}$ .
- The digital output - **No** - is full scale (03FFH) (1023 decimal) when the input signal is equal to, or higher than,  $V_{+ref}$ , and zero when the input signal is equal to, or lower than  $V_{-ref}$ .
- Conversion results may be in straight binary format or 2s-complement format. The conversion formula for the ADC result when using straight binary format is

$$n = \left\lfloor \frac{(V_{in} - V_{-ref}) 2^N}{V_{+ref} - V_{-ref}} + 1/2 \right\rfloor$$

Or with  $V_{-ref} = 0V$ , and  $V_{+ref} = V_{REF}$  :

$$n = \left\lfloor \frac{(V_{in}) 2^N}{V_{+ref}} + 1/2 \right\rfloor$$

**NOTE: Don't forget to round the result to the nearest integer**

- ADC10 operation is governed by user-written software. The user must specify all options via the ADC10 control registers.
- ADC10 is configured by two 16-bit control registers, ADC10CTL0 and ADC10CTL1. **All bits** of both registers **must** be specified by the user.
- **The convertor is enabled with the ADC10ON bit. With few exceptions the ADC10 control bits can only be modified when ENC = 0. ENC must however be set to 1 before any conversion can take place.**
- An additional, third register - **ADC10AE** - enables external port pins to be used as inputs to the ADC

**Exercises:**

A 10-bit ADC has a reference voltage of 3.3V ( $V_{+ref} = 3.3V$ ,  $V_{-ref} = 0V$ )

Q1

What is the digital output for each of the following input voltages?

- (i)  $V_i = 3.0 V$
- (ii)  $V_i = 1.5V$

Q2

What is the analogue input voltage if the observed digital output is;

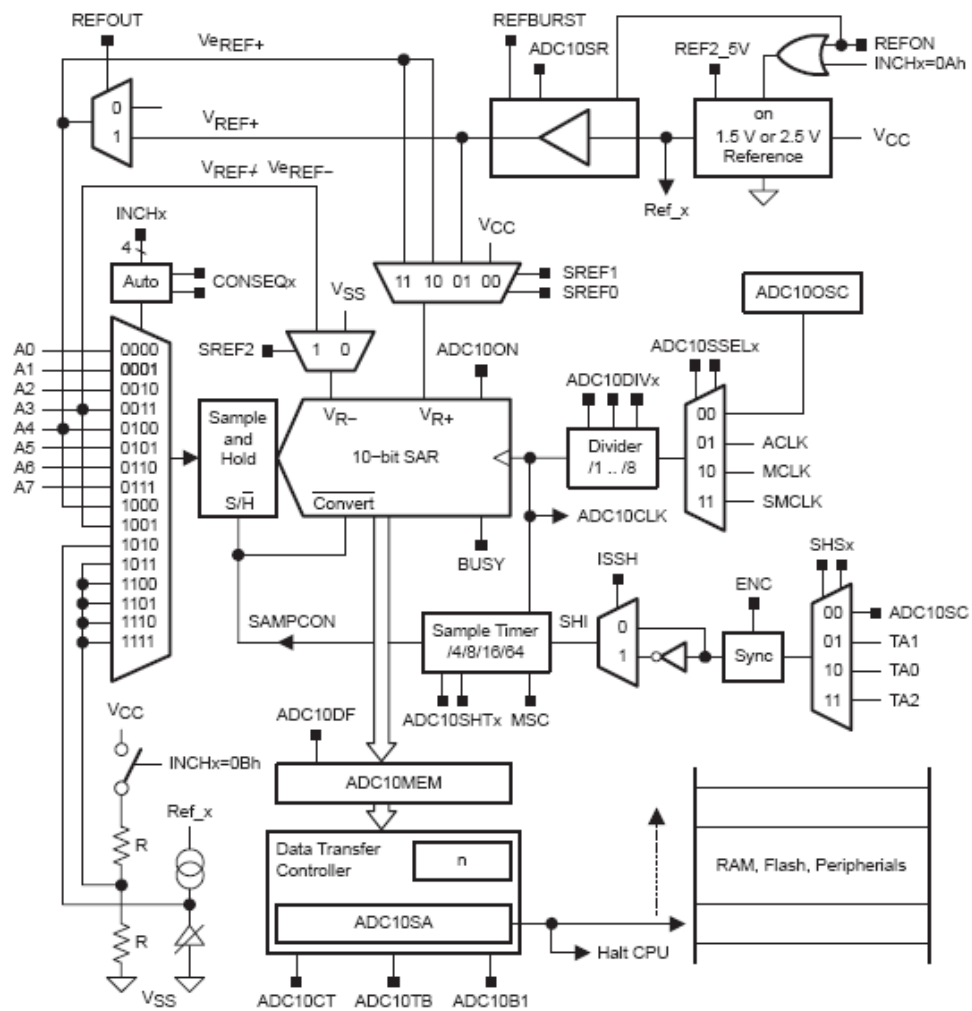
- (i) 1000 decimal
- (ii) 1FF Hex

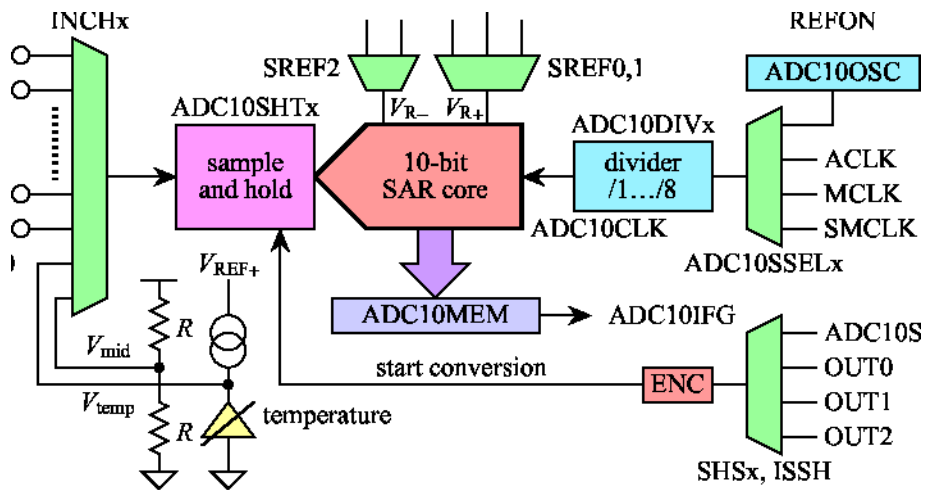
Q3

What is the smallest change in input voltage, guaranteed to cause a change in the digital output (i.e what is the Quantization size of the ADC)?

The converter and its associated auxiliary circuitry (from G2553 datasheet) is represented in the block diagram below. A more stylized version is given overleaf.

Figure 18-1. ADC10 Block Diagram





**Figure: Stylised version of ADC10 as implemented on G2553**

**Operation of ADC10** : The module is configured/controlled by user software. The main registers governing converter operation are (i) **ADC10CTL0** and (ii) **ADC10CTL1**. Both registers are 16-bit. **All bits** must be specified by the user. Using these registers the user selects the signal to be converted, the clock of the convertor, the clock divisor, the convertor reference voltages, the binary output format, etc etc of ADC10. An additional, third register **ADC10AE** enables external port pins to be used as inputs to the ADC

We will examine each register in turn;

**ADC10CTL0 – 16 bits**

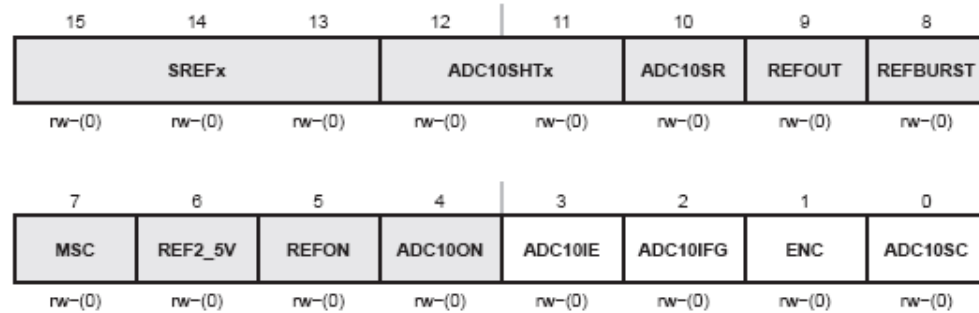
**ADC10CTL1 – 16 bits**

**ADC10AE – 8-bits, used for external analog input signals on P1.0 – P1.7**

**Assume that we want to configure the ADC to operate as follows;**

- *Operate on the internal MSP430 clock – SMCLK. This is the default clock available*
- *Do single conversions on A0 (analog signal applied on P1.0)*
- *Have reference voltage equal to the MSP430 supply (3.3V nominally)*
- *Produce a simple 10-bit binary output – 0 to 1023*
- *Minimal S/H time*
- *No interrupts used*
- *Conversion started by setting SC bit on ADC10CTL0*
- *EOC (End of Conversion) detected by polling – waiting for ADC10BUSY bit on ADC10CTL1 to go LOW (it remains HIGH while conversion is ongoing)*

## ADC10CTL0, ADC10 Control Register 0



Modifiable only when ENC = 0

|               |               |                                                                                                                                                                                     |                                                        |
|---------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| SREFx         | Bits<br>15-13 | Select reference                                                                                                                                                                    |                                                        |
|               |               | 000                                                                                                                                                                                 | $V_{R+} = V_{CC}$ and $V_{R-} = V_{SS}$                |
|               |               | 001                                                                                                                                                                                 | $V_{R+} = V_{REF+}$ and $V_{R-} = V_{SS}$              |
|               |               | 010                                                                                                                                                                                 | $V_{R+} = V_{REF+}$ and $V_{R-} = V_{SS}$              |
|               |               | 011                                                                                                                                                                                 | $V_{R+} = V_{REF+}$ and $V_{R-} = V_{SS}$              |
|               |               | 100                                                                                                                                                                                 | $V_{R+} = V_{CC}$ and $V_{R-} = V_{REF+} / V_{REF-}$   |
|               |               | 101                                                                                                                                                                                 | $V_{R+} = V_{REF+}$ and $V_{R-} = V_{REF+} / V_{REF-}$ |
|               |               | 110                                                                                                                                                                                 | $V_{R+} = V_{REF+}$ and $V_{R-} = V_{REF+} / V_{REF-}$ |
| ADC10<br>SHTx | Bits<br>12-11 | ADC10 sample-and-hold time                                                                                                                                                          |                                                        |
|               |               | 00                                                                                                                                                                                  | 4 x ADC10CLKs                                          |
|               |               | 01                                                                                                                                                                                  | 8 x ADC10CLKs                                          |
|               |               | 10                                                                                                                                                                                  | 16 x ADC10CLKs                                         |
|               |               | 11                                                                                                                                                                                  | 64 x ADC10CLKs                                         |
| ADC10SR       | Bit 10        | ADC10 sampling rate. This bit selects the reference buffer drive capability for the maximum sampling rate. Setting ADC10SR reduces the current consumption of the reference buffer. |                                                        |
|               |               | 0                                                                                                                                                                                   | Reference buffer supports up to ~200 ksp/s             |
|               |               | 1                                                                                                                                                                                   | Reference buffer supports up to ~50 ksp/s              |
| REFOUT        | Bit 9         | Reference output                                                                                                                                                                    |                                                        |
|               |               | 0                                                                                                                                                                                   | Reference output off                                   |
|               |               | 1                                                                                                                                                                                   | Reference output on                                    |
| REFBURST      | Bit 8         | Reference burst. REFOUT must also be set.                                                                                                                                           |                                                        |
|               |               | 0                                                                                                                                                                                   | Reference buffer on continuously                       |
|               |               | 1                                                                                                                                                                                   | Reference buffer on only during sample-and-conversion  |

|                 |       |                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MSC</b>      | Bit 7 | Multiple sample and conversion. Valid only for sequence or repeated modes.<br>0 The sampling requires a rising edge of the SHI signal to trigger each sample-and-conversion.<br>1 The first rising edge of the SHI signal triggers the sampling timer, but further sample-and-conversions are performed automatically as soon as the prior conversion is completed |
| <b>REF2_5V</b>  | Bit 6 | Reference-generator voltage. REFON must also be set.<br>0 1.5 V<br>1 2.5 V                                                                                                                                                                                                                                                                                         |
| <b>REFON</b>    | Bit 5 | Reference generator on<br>0 Reference off<br>1 Reference on                                                                                                                                                                                                                                                                                                        |
| <b>ADC10ON</b>  | Bit 4 | ADC10 on<br>0 ADC10 off<br>1 ADC10 on                                                                                                                                                                                                                                                                                                                              |
| <b>ADC10IE</b>  | Bit 3 | ADC10 interrupt enable<br>0 Interrupt disabled<br>1 interrupt enabled                                                                                                                                                                                                                                                                                              |
| <b>ADC10IFG</b> | Bit 2 | ADC10 interrupt flag. This bit is set if ADC10MEM is loaded with a conversion result. It is automatically reset when the interrupt request is accepted, or it may be reset by software. When using the DTC this flag is set when a block of transfers is completed.<br>0 No interrupt pending<br>1 Interrupt pending                                               |
| <b>ENC</b>      | Bit 1 | Enable conversion<br>0 ADC10 disabled<br>1 ADC10 enabled                                                                                                                                                                                                                                                                                                           |
| <b>ADC10SC</b>  | Bit 0 | Start conversion. Software-controlled sample-and-conversion start. ADC10SC and ENC may be set together with one instruction. ADC10SC is reset automatically.<br>0 No sample-and-conversion start<br>1 Start sample-and-conversion                                                                                                                                  |

## Discussion of ADC10CTL0 bits

- Properties (bits) colored green are left to their default values (0). You will rarely (ever) have to consider anything other than the default value(s).
- Properties (bits) colored black will need to be carefully considered and precise values chosen, depending on the application and desired operation mode of the ADC.

### SREFx bits:

These bits control the choice of reference voltage for the ADC10 module and thus the range of applicable input voltage.

Many options are available, *the simplest being 000 which sets  $V_{R+}$  to 3.3V and  $V_{R-}$  to 0V (gnd). This means that an applied analog input voltage may vary from ground up to 3.3V.*

**Alternatively use may be made of internal reference voltages, 2.5V being the most common etc etc.**

Ultimately the choice of reference will depend on the range of the analog signal being converted. **Usually option 000 or 001 is used here**

| SREFx | Bits  | Select reference                                       |
|-------|-------|--------------------------------------------------------|
|       | 15-13 |                                                        |
|       | 000   | $V_{R+} = V_{CC}$ and $V_{R-} = V_{SS}$                |
|       | 001   | $V_{R+} = V_{REF+}$ and $V_{R-} = V_{SS}$              |
|       | 010   | $V_{R+} = V_{REF+}$ and $V_{R-} = V_{SS}$              |
|       | 011   | $V_{R+} = V_{REF+}$ and $V_{R-} = V_{SS}$              |
|       | 100   | $V_{R+} = V_{CC}$ and $V_{R-} = V_{REF-} / V_{REF+}$   |
|       | 101   | $V_{R+} = V_{REF+}$ and $V_{R-} = V_{REF-} / V_{REF+}$ |
|       | 110   | $V_{R+} = V_{REF+}$ and $V_{R-} = V_{REF-} / V_{REF+}$ |
|       | 111   | $V_{R+} = V_{REF+}$ and $V_{R-} = V_{REF-} / V_{REF+}$ |

### ADC10SHTx bits:

This governs the number of CLOCK cycles for which the Sample and Hold is activated prior to conversion

| ADC10 SHTx | Bits  | ADC10 sample-and-hold time |
|------------|-------|----------------------------|
|            | 12-11 |                            |
|            | 00    | 4 x ADC10CLKs              |
|            | 01    | 8 x ADC10CLKs              |
|            | 10    | 16 x ADC10CLKs             |
|            | 11    | 64 x ADC10CLKs             |

**00** holds for the shortest time and thus allows the fastest conversion rate.

### ADC SR bit: Selects the sampling rate maximum

| ADC10SR | Bit 10 | ADC10 sampling rate. This bit selects the reference buffer drive capability for the maximum sampling rate. Setting ADC10SR reduces the current consumption of the reference buffer. |
|---------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | 0      | Reference buffer supports up to ~200 ksp/s                                                                                                                                          |
|         | 1      | Reference buffer supports up to ~50 ksp/s                                                                                                                                           |

For our purposes this value is not crucial - we will assign a value of **0**

**REFOUT bit:** Decides whether or not the reference voltage used in ADC10 will be available to external circuitry via a port pin.

|        |       |                      |
|--------|-------|----------------------|
| REFOUT | Bit 9 | Reference output     |
|        | 0     | Reference output off |
|        | 1     | Reference output on  |

We will conserve port pins for input/output and thus assign **0** to this bit.

**REFBURST bit:** Irrelevant if REFOUT is reset

|          |       |                                                       |
|----------|-------|-------------------------------------------------------|
| REFBURST | Bit 8 | Reference burst. REFOUT must also be set.             |
|          | 0     | Reference buffer on continuously                      |
|          | 1     | Reference buffer on only during sample-and-conversion |

We will thus assign **0** to this bit.

**MSC BIT** We will do one conversion at a time and thus give this bit a value **0**

|     |       |                                                                                                                                                                                  |
|-----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MSC | Bit 7 | Multiple sample and conversion. Valid only for sequence or repeated modes.                                                                                                       |
|     | 0     | The sampling requires a rising edge of the SHI signal to trigger each sample-and-conversion.                                                                                     |
|     | 1     | The first rising edge of the SHI signal triggers the sampling timer, but further sample-and-conversions are performed automatically as soon as the prior conversion is completed |

**REF2.5V bit:** Allows you to choose a 2.5V or 1.5V reference (refer to SREFx bits), if an internal reference voltage is used.

|         |       |                                                      |
|---------|-------|------------------------------------------------------|
| REF2_5V | Bit 6 | Reference-generator voltage. REFON must also be set. |
|         | 0     | 1.5 V                                                |
|         | 1     | 2.5 V                                                |

We will thus assign **0** to this bit – i.e we will use the 3.3V supply as a reference.

**REFON bit :** Turn on reference voltage generator, if using it

|       |       |                        |
|-------|-------|------------------------|
| REFON | Bit 5 | Reference generator on |
|       | 0     | Reference off          |
|       | 1     | Reference on           |

We will thus assign **0** to this bit.

**ADC10ON bit:** Turn on ADC10

|         |       |           |
|---------|-------|-----------|
| ADC10ON | Bit 4 | ADC10 on  |
|         | 0     | ADC10 off |
|         | 1     | ADC10 on  |

We will assign **1** to this bit.

### ADC10IE bit : Interrupt Enable

|         |       |                        |
|---------|-------|------------------------|
| ADC10IE | Bit 3 | ADC10 interrupt enable |
|         | 0     | Interrupt disabled     |
|         | 1     | interrupt enabled      |

We do not, for the moment, make use of interrupts and will thus assign **0** to this bit

### ADC10IFG bit: Interrupt flag bit

|          |       |                                                                                                                                                                                                                                                                     |
|----------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADC10IFG | Bit 2 | ADC10 interrupt flag. This bit is set if ADC10MEM is loaded with a conversion result. It is automatically reset when the interrupt request is accepted, or it may be reset by software. When using the DTC this flag is set when a block of transfers is completed. |
|          | 0     | No interrupt pending                                                                                                                                                                                                                                                |
|          | 1     | Interrupt pending                                                                                                                                                                                                                                                   |

This bit is set/reset by ADC10 itself. We can set it to any value – **0** in this case

**ENC bit** : Enable Conversion. With few exceptions *the ADC10 control bits can only be modified when ENC = 0. ENC must then be set to 1 before any conversion can take place.*

*WE normally keep this bit at 0 while setting the ADC10 registers and then change it to 1 before starting conversions*

|     |       |                   |
|-----|-------|-------------------|
| ENC | Bit 1 | Enable conversion |
|     | 0     | ADC10 disabled    |
|     | 1     | ADC10 enabled     |

*Effectively ENC acts as a lock on the ADC10 control registers (with the exception of a few bits).*

- *With ENC =0 the two registers (ADCCTLO and ADCCTL1) CONTENTS CAN BE CHANGED.*
- *With ENC = 1 the two registers (ADCCTLO and ADCCTL1)contents CANNOT BE CHANGED.*

*We write a 0 here initially at least – disable ADC10 while configuring it. This will be changed to a 1 before using the ADC.*

### ADC10SC bit: ADC10 Start Conversion bit

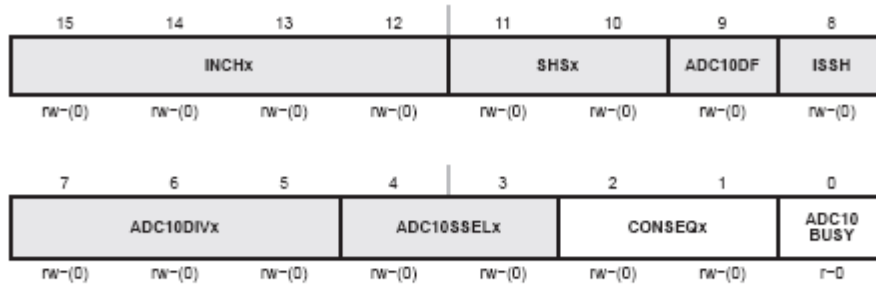
|         |       |                                                                                                                                                              |
|---------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADC10SC | Bit 0 | Start conversion. Software-controlled sample-and-conversion start. ADC10SC and ENC may be set together with one instruction. ADC10SC is reset automatically. |
|         | 0     | No sample-and-conversion start                                                                                                                               |
|         | 1     | Start sample-and-conversion                                                                                                                                  |

Set this bit to one whenever a conversion is required. Leave it at zero when control registers are being set up

For our purposes a suitable load value to **ADCCTL0** is **0x0010**, as discussed above. Remember that **ENC** must be set before we attempt any conversions.

## DATSHEET Extract – G2553

**ADC10CTL1, ADC10 Control Register 1**



Modifiable only when ENC = 0

|                |               |                                                                                                                                   |                                          |
|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <b>INCHx</b>   | Bits<br>15-12 | Input channel select. These bits select the channel for a single-conversion or the highest channel for a sequence of conversions. |                                          |
|                |               | 0000                                                                                                                              | A0                                       |
|                |               | 0001                                                                                                                              | A1                                       |
|                |               | 0010                                                                                                                              | A2                                       |
|                |               | 0011                                                                                                                              | A3                                       |
|                |               | 0100                                                                                                                              | A4                                       |
|                |               | 0101                                                                                                                              | A5                                       |
|                |               | 0110                                                                                                                              | A6                                       |
|                |               | 0111                                                                                                                              | A7                                       |
|                |               | 1000                                                                                                                              | VeREF+                                   |
|                |               | 1001                                                                                                                              | VREF-/VeREF-                             |
|                |               | 1010                                                                                                                              | Temperature sensor                       |
|                |               | 1011                                                                                                                              | $(V_{CC} - V_{SS}) / 2$                  |
|                |               | 1100                                                                                                                              | $(V_{CC} - V_{SS}) / 2$                  |
|                |               | 1101                                                                                                                              | $(V_{CC} - V_{SS}) / 2$                  |
|                |               | 1110                                                                                                                              | $(V_{CC} - V_{SS}) / 2$                  |
|                |               | 1111                                                                                                                              | $(V_{CC} - V_{SS}) / 2$                  |
| <b>SHSx</b>    | Bits<br>11-10 | Sample-and-hold source select                                                                                                     |                                          |
|                |               | 00                                                                                                                                | ADC10SC bit                              |
|                |               | 01                                                                                                                                | Timer_A.OUT1                             |
|                |               | 10                                                                                                                                | Timer_A.OUT0                             |
|                |               | 11                                                                                                                                | Timer_A.OUT2                             |
| <b>ADC10DF</b> | Bit 9         | ADC10 data format                                                                                                                 |                                          |
|                |               | 0                                                                                                                                 | Straight binary                          |
|                |               | 1                                                                                                                                 | 2's complement                           |
| <b>ISSH</b>    | Bit 8         | Invert signal sample-and-hold                                                                                                     |                                          |
|                |               | 0                                                                                                                                 | The sample-input signal is not inverted. |
|                |               | 1                                                                                                                                 | The sample-input signal is inverted.     |

|                        |             |                                                                         |                                              |
|------------------------|-------------|-------------------------------------------------------------------------|----------------------------------------------|
| <b>ADC10DIVx</b>       | Bits<br>7-5 | ADC10 clock divider                                                     |                                              |
|                        |             | 000                                                                     | /1                                           |
|                        |             | 001                                                                     | /2                                           |
|                        |             | 010                                                                     | /3                                           |
|                        |             | 011                                                                     | /4                                           |
|                        |             | 100                                                                     | /5                                           |
|                        |             | 101                                                                     | /8                                           |
|                        |             | 110                                                                     | /7                                           |
|                        |             | 111                                                                     | /8                                           |
| <b>ADC10<br/>SSELx</b> | Bits<br>4-3 | ADC10 clock source select                                               |                                              |
|                        |             | 00                                                                      | ADC10OSC                                     |
|                        |             | 01                                                                      | ACLK                                         |
|                        |             | 10                                                                      | MCLK                                         |
|                        |             | 11                                                                      | SMCLK                                        |
| <b>CONSEQx</b>         | Bits<br>2-1 | Conversion sequence mode select                                         |                                              |
|                        |             | 00                                                                      | Single-channel-single-conversion             |
|                        |             | 01                                                                      | Sequence-of-channels                         |
|                        |             | 10                                                                      | Repeat-single-channel                        |
|                        |             | 11                                                                      | Repeat-sequence-of-channels                  |
| <b>ADC10<br/>BUSY</b>  | Bit 0       | ADC10 busy. This bit indicates an active sample or conversion operation |                                              |
|                        |             | 0                                                                       | No operation is active.                      |
|                        |             | 1                                                                       | A sequence, sample, or conversion is active. |

## Discussion of ADC10CTL1 bits

- Properties (bits) colored green are left to their default values (0). You will rarely (ever) have to consider anything other than the default value(s).
- Properties (bits) colored black will need to be carefully considered and precise values chosen, depending on the application and desired operation mode of the ADC.

### INCHx bits:

These bits decide which analog voltage is passed through the multiplexer to be converted by ADC10. Note that 8 external analog voltages may be connected as well as the three useful internal signals – (i) Temperature sensor output on channel 10, (ii) reference voltage on channel 8 and (iii) a voltage related to the supply voltage of the G2553 on several channels. This voltage is used to monitor the G2553 supply when it is battery-powered.

| INCHx | Bits  | Input channel select. These bits select the channel for a single-conversion or the highest channel for a sequence of conversions. |
|-------|-------|-----------------------------------------------------------------------------------------------------------------------------------|
|       | 15-12 |                                                                                                                                   |
|       |       | 0000 A0                                                                                                                           |
|       |       | 0001 A1                                                                                                                           |
|       |       | 0010 A2                                                                                                                           |
|       |       | 0011 A3                                                                                                                           |
|       |       | 0100 A4                                                                                                                           |
|       |       | 0101 A5                                                                                                                           |
|       |       | 0110 A6                                                                                                                           |
|       |       | 0111 A7                                                                                                                           |
|       |       | 1000 V <sub>REF+</sub>                                                                                                            |
|       |       | 1001 V <sub>REF-</sub> /V <sub>REF-</sub>                                                                                         |
|       |       | 1010 Temperature sensor                                                                                                           |
|       |       | 1011 (V <sub>CC</sub> - V <sub>SS</sub> ) / 2                                                                                     |
|       |       | 1100 (V <sub>CC</sub> - V <sub>SS</sub> ) / 2                                                                                     |
|       |       | 1101 (V <sub>CC</sub> - V <sub>SS</sub> ) / 2                                                                                     |
|       |       | 1110 (V <sub>CC</sub> - V <sub>SS</sub> ) / 2                                                                                     |
|       |       | 1111 (V <sub>CC</sub> - V <sub>SS</sub> ) / 2                                                                                     |

For the analog input channel A0select **0000** or channel 0

### SHSx bits:

This governs how a conversion is started – either by the on-board timer or by setting the ADC10SC bit. We will use the latter – set these bits to **00**

| SHSx | Bits  | Sample-and-hold source select |
|------|-------|-------------------------------|
|      | 11-10 |                               |
|      |       | 00 ADC10SC bit                |
|      |       | 01 Timer_A.OUT1               |
|      |       | 10 Timer_A.OUT0               |
|      |       | 11 Timer_A.OUT2               |

-----

**ADC10DF bit:** Selects the format of the conversion result.

| ADC10DF | Bit 9 | ADC10 data format |
|---------|-------|-------------------|
|         |       | 0 Straight binary |
|         |       | 1 2's complement  |

For our purposes, we will generally require a straightforward binary answer - we will assign a value of **0**

**ISSH bit:** Invert SH signal – value not critical for our purposes – assign a **0**

|             |              |                                          |
|-------------|--------------|------------------------------------------|
| <b>ISSH</b> | <b>Bit 8</b> | Invert signal sample-and-hold            |
|             | 0            | The sample-input signal is not inverted. |
|             | 1            | The sample-input signal is inverted.     |

**ADC10DIVx bit:** ADC10 clock divider bits. For fastest conversions divide by 1 i.e reset bits to **000**

|                  |             |                            |
|------------------|-------------|----------------------------|
| <b>ADC10DIVx</b> | <b>Bits</b> | <b>ADC10 clock divider</b> |
|                  | 7-5         | 000 /1                     |
|                  |             | 001 /2                     |
|                  |             | 010 /3                     |
|                  |             | 011 /4                     |
|                  |             | 100 /5                     |
|                  |             | 101 /6                     |
|                  |             | 110 /7                     |
|                  |             | 111 /8                     |

**ADC10SSELx bit:** ADC10 clock selection bits. The internal oscillator or any of the three standard G2553 clocks may be selected – we will use the Sub-system Master Clock or SMCLK – use **11** for these bits.

|                   |             |                                  |
|-------------------|-------------|----------------------------------|
| <b>ADC10SSELx</b> | <b>Bits</b> | <b>ADC10 clock source select</b> |
|                   | 4-3         | 00 ADC10OSC                      |
|                   |             | 01 ACLK                          |
|                   |             | 10 MCLK                          |
|                   |             | 11 SMCLK                         |

**COCSEQx bits** Govern how many conversions are attempted

|                |             |                                        |
|----------------|-------------|----------------------------------------|
| <b>CONSEQx</b> | <b>Bits</b> | <b>Conversion sequence mode select</b> |
|                | 2-1         | 00 Single-channel-single-conversion    |
|                |             | 01 Sequence-of-channels                |
|                |             | 10 Repeat-single-channel               |
|                |             | 11 Repeat-sequence-of-channels         |

We will convert one channel at a time – use **00** for these bits

**ADC10BUSY bit:** Set/reset by ADC10 to indicate whether or not a conversion is in progress. The value written by the user is irrelevant – we will reset to **0**

|                  |              |                                                                                |
|------------------|--------------|--------------------------------------------------------------------------------|
| <b>ADC10BUSY</b> | <b>Bit 0</b> | <b>ADC10 busy. This bit indicates an active sample or conversion operation</b> |
|                  | 0            | No operation is active.                                                        |
|                  | 1            | A sequence, sample, or conversion is active.                                   |

For our purposes a suitable load value to **ADCCTL1** is **0x0018**, as discussed above.

## **CODE Segments for ADC10 Control**

### ➤ Initialization code for ADC10

```
//Code to Control ADC10
// SET ENC to 0 (bit 1 – 2nd bit -of ADC10CTL0 )so that we can change register values
ADC10CTL0= ADC10CTL0 & 0x????; without altering any other bits in register

// Set up ADC – Analog i/p AO (Channel 0) is i/p to ADC
ADC10CTL1 = 0x????; // Control register 1
ADC10CTL0 = 0x????; // Control register 0

/// Now set the ENC bit on ADC10CTL0 to 1 without altering any other bits in register

ADC10CTL0= ADC10CTL0 | 0x????;
```

### ➤ Starting a Conversion on ADC10:

Look at the datasheet or through the notes and work out what value should be placed in **ADC10SC** bit of the 16-bit **ADC10CTL0** register to start a conversion and fill in the question marks.

Thus, to start a conversion on the (correctly configured) ADC10 the following line may be used:

```
// Start a conversion
ADC10CTL0 = ADC10CTL0 | 0x????;
```

➤ **Waiting for (recognizing) the End of Conversion;**

Look at the notes or the datasheet to find out which bit of the ADC10CTL1 register should be high when a conversion is in progress. When can check that this bit to see if its low to let us know a conversion is complete. Fill in the ???? in the code below:

Thus a simple *wait until end of conversion* process can be achieved using the C commands:

```
while((ADC10CTL1&0x????)==?)    // allow conversion to complete
{
}
```

➤ **Reading conversion result (At last!);**

The final result may be read, after conversion has completed, from the 10-bit ADC10MEM register. The result will be a 10-bit number, so use an integer variable (**not** a char) to hold it. The value will be somewhere in the range 0 to 1023. Fill in the question marks.

```
int result;                // integer variable for 10-bit result
// program to configure and start conversion etc
//
// program to configure and start conversion etc
//
//
result = ???;              // Get the result
//
```

***Note on use of external analog signals as inputs to ADC10.***

- Apply analog signals (with correct range!!) to appropriate input pins – refer to pinout of G2253 given below. Analog inputs are marked as A0, A1 etc.
- Enable analog inputs on the pins used via the ADC10AE0 register, as detailed below

**ADC10AE0: Analog Input control Register**

ADC10AE, Analog (input) Enable Control Register



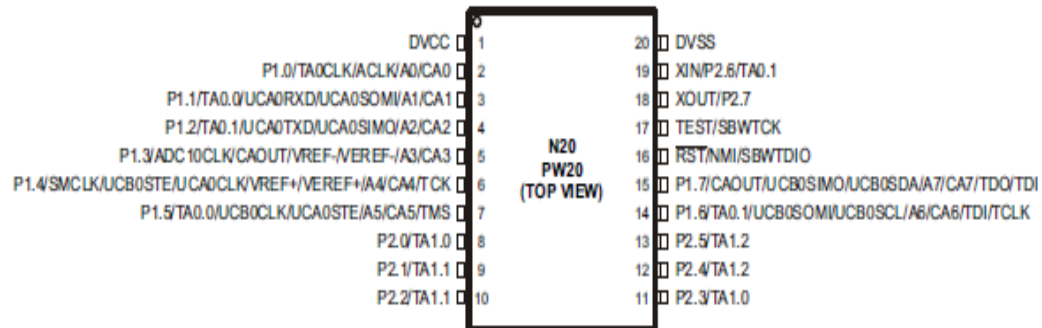
|                      |      |                         |
|----------------------|------|-------------------------|
| ADC10AE <sub>x</sub> | Bits | ADC10 analog enable     |
|                      | 7-0  | 0 Analog input disabled |
|                      |      | 1 Analog input enabled  |

Thus at microcontroller initialization, the port pin on which an analog voltage is to be read must be configured as an input (via PxDIR) and then analog signals must be enabled on that port pin via ADC10AE0

Look at the notes or data sheet to work out which port and pin connects to A0  
Thus to allow an analog input signal on A0 ( i.e. on P1.0, refer to pinout overleaf)  
the following Port setup could be used

```
P?DIR=0x00 // make P?.0 (A0) an input - an analog input!  
P?SEL=0x00  
P?IE=0x00;  
ADC10AE0 = ????; // allow analog signal input on A0
```

## Device Pinout, MSP430G2x13 and MSP430G2x53, 20-Pin Devices, TSSOP and PDIP



## MSP430-G2553 Pin-out

The pin-out of the DIP version is given above.